# Supplementary material: Quantum and classical physics-informed networks for laser-pulse thermoelasticity

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## S1 Reproducibility note and evidence inventory

The internally prespecified brief posed descriptive research questions without formal statistical hypotheses or equivalence margins. Execution expanded the learning-rate grid, added L-BFGS, and used exact outer-boundary traces and a near-surface exclusion for both initial-rate penalties. E1b adds retuned classical comparisons after identifying inert circuit coordinates. Table S1 identifies the complete evidence: all earlier per-run records are unchanged, and all retained E3 and secondary-wave-speed cells contain their declared seeds.

The versioned archive is `QPINN_LTE_release_v1_2026-10-21.zip`. Its analysis directory contains `qpinn_local.py`, `ground_truth.py`, `noise_eval.py`, `run_e4.py`, all launchers, `select_lrs.py`, `functional_count.py`, `family_equivalence.py`, the four exact-solution verification scripts, and the table, figure, and record generators. The results directory contains the 440 retained per-run JSON records collected in `runs_all.json` and 24 further JSON files of other types: 10 segregated `*_superseded_*` per-run records from discarded launches (used by no table), the current and superseded E2/E4 aggregate files, three selected-rate summaries, and seven pilot and diagnostic files (464 JSON files in all). It also contains the trained checkpoint state dictionaries (456 `.pt` files, 6.9 MB of content: 440 retained, 10 superseded, 6 diagnostic), full-precision E2 aggregate records in `e2_exp_g9.json`, and corrected, checkpoint-labeled E4 records in `e4_noise.json`. The logs directory (527 files, 0.45 MB) includes execution logs, queue dispatcher outputs, and the initial-rate weight-ablation logs and variants. The package contains the tables and figures, `runs_all.json` with training histories and closure counts, verification log, coordinate-audit output, family-equivalence check, checkpoint gains, and selected rates. `RELEASE_README` maps every manuscript table and figure to its input files and generator and gives regeneration commands and launcher thread settings. Table S1 separates the retained analysis records from the provenance-only launches. The archive E2 file corresponds to the package input `e2_gradvar.json`; individual E2 gradient draws were not retained and are distinct from these aggregate records. Repository address/DOI: *to be inserted by the author at submission.*

`results_package_v6.md` is the current executed-results description; its numerical tables are unchanged. Its protocol prose and captions use the single-seed initial-rate ablation values, distinguish the smallest searched spectral width from the architecture's possible minimum, give comparator deviations separately, and avoid inferring representational impossibility or unchanged gradient-variance decay. The corrected `closed_form_appendix.md` and `ground_truth.py` header separate the pulse-specific finite-difference maxima, printed inverse-transform agreement, smooth-pulse discrepancy, and stored-reference regression check, consistently with Table S15. All are pointwise checks, not a uniform accuracy bound.

The E4 records use the corrected stress-output mapping; the earlier incorrect evaluator outputs are excluded. The table generator groups by pulse, wave speed, circuit, channel probability, and shot budget. Clean controls share the noise grid and checkpoints. The verification-code comments

distinguish zero initial elastic-strain rate from nonzero initial stress rate. The selected readout gains reported here come from the checkpoint state-dictionary audit in the archive.

Initial E3 and secondary-speed jobs with hard-coded rates were discarded before use, and their outputs are segregated. Completed physics sweeps transfer the E1 exponential-pulse rates; new circuit architectures were not independently retuned. Data-only rates were selected separately. The circuit collocation jobs at $N_c = 4096$ were stopped and dropped for cost; this is a declared deviation from the plan (main Section 4.3). The auxiliary dispatcher skips a job when its log exists, so completion was checked using the actual per-cell records, not its end marker. Its adjustable worker file and overlapping queues prevent treating all recorded durations as comparable throughput.

Table S1: Final evidence inventory. Counts denote records, checkpoint files, or reported individual runs, not independent experiments across all rows. E1b reuses E1 circuit results and repeats the smaller spectral model. E3 repeats the E1 settings for its anchor configurations and repeats the classical architectures associated with the two nominal-count-84 circuit rows.

| Source/group | Count | Content |
|---|---|---|
| `runs_all.json`: e1 | 105 | Main production runs |
| fmatch | 60 | Classical E1b production runs |
| ablation_xmin | 9 | Five reference, two MLP, two circuit runs |
| ablation_repro | 1 | Reference replay at E1 thread setting |
| e3 | 66 | 36 parameter, 14 collocation, 16 label runs |
| gamma | 24 | Two speeds, four models, three seeds |
| lrsel | 105 | E1 learning-rate candidates |
| lrsel_fmatch | 60 | E1b candidates |
| lrsel_e3_data | 10 | Data-only candidates |
| Total in `runs_all.json` | 440 | Histories and closure counts retained |
| Superseded per-run records | 10 | Discarded launches, segregated; used by no table |
| Other JSON files | 14 | E2/E4 aggregates (2 current, 2 superseded), 3 selected-rate summaries, 7 pilot/diagnostic files |
| Total JSON files | 464 | 440 retained records plus the 24 files above |
| Checkpoint state dictionaries | 456 | `.pt`: 440 retained, 10 superseded, 6 diagnostic; 6.9 MB |
| Execution logs | 527 | files, 0.45 MB; includes dispatcher and ablation logs |
| `e2_gradvar.json` | 81 | Aggregate variance cells |
| Corrected `e4_noise.json` | 144 | Selected-checkpoint conditions |
| Initial-rate weight ablation | 6 | Log-derived, not in `runs_all.json` |

The reference-MLP cutoff-0.3 ablation used a different thread setting from E1. Its recorded Adam history is identical, but its L-BFGS endpoint is not. The new replay at the E1 setting reproduces all recorded numerical histories, errors, validation values, and closure counts exactly; elapsed time and record identifiers naturally differ (main Table 4). The E3 anchor reruns and repeated classical architectures also reproduce their numerical outcomes. Some anchor reruns used the auxiliary queue's thread setting, demonstrating that the observed thread dependence is not universal across the tested models (Table S3; main Table 4). These checks establish reproducibility of the reported repeats, not a guarantee of bitwise determinism for every run or platform.

The reference ablation has stress error 0.0496 after 1060 closures with 2 threads; the E1 run and its replay give 0.0461 after 1065 closures with 3 threads (main Table 4). Twelve E3 anchor records (two configurations, three families, two seeds) reproduce their E1 numerical histories and outcomes. The classical networks associated with the two nominal-count-84 targets also repeat their outcomes (main Section 4.3 and the E3 per-seed tables). These repeated executions are not extra independent seeds.

A plausible explanation is that intra-op parallel reductions become relevant only for sufficiently large tensor operations. The reference network's fixed L-BFGS batch has 2048 points and width 64, whereas its Adam batch has 512 points (main Tables 2–3). A thread-dependent floating-point summation order could then create small gradient changes that the strong-Wolfe search amplifies into a different endpoint. This interpretation is consistent with the phase-specific discrepancy and smaller-model repetitions, but the operation-level reduction policy, its grain-size threshold, and the amplification mechanism were not separately instrumented or tested. The source declarations below are not per-run measurements of effective thread use.

## S2 Parameter accounting

Table S2 gives circuit nominal and reported output-affecting counts and the nearest nominal-count classical designs. Signed deviation is $100(P_{\text{classical}}/P_{\text{target}}-1)$. For classical models the displayed output count equals the nominal count under the convention that no coordinates have been structurally removed; this is not an empirical rank claim. The released `functional_count.py` audits every registered scalar at model-initialization seed 0, with 64 input pairs drawn uniformly from $[-1,1]^2$ using a separate input generator seeded with 12345. Each scalar is changed by $+0.37$ while all others are held fixed and then restored; a coordinate is classified as inert if the largest absolute change of either output is below $10^{-12}$. The script's executed output, `functional_parameter_audit.txt`, reports every inert layer/qubit coordinate and the totals in Table S2. Layer and qubit indices in that file start at zero. The test includes encoding scales, rotation angles, and affine readout parameters; the perturbation is an angle increment only for the rotation coordinates. No readout coordinate is classified as inert. Individual numerical changes are not printed by this script. These counts describe output-affecting coordinates at the tested initialization and inputs, not a measured Jacobian rank, intrinsic function-space dimension, or proof that all remaining coordinates are independent.

For $L = 2$, the 4-, 6-, and 8-qubit circuits realize the same output function family under the qubit relabeling $\{0,1,2,3\} \leftrightarrow \{0,1,2,n_q-1\}$. The last CZ ring commutes with each measured $Z$ and cancels in its expectation. Conjugation through the preceding CZ ring couples the measured qubits 0 and 1 only to their ring neighbors 2 and $n_q-1$; all other local gates cancel from those expectations. This leaves the same active-coordinate dependence for each width, with the 30 output-affecting coordinates in Table S2. The executed `family_equivalence.py` check transplants encoding scales, rotation angles, and affine readouts of the 4-qubit circuit onto these qubits of the wider circuits. On 64 random input pairs, `family_equivalence_check.txt` reports maximum absolute raw-output differences of $5.55 \times 10^{-16}$ and $8.88 \times 10^{-16}$ for 6 and 8 qubits. The same transplantation at $L = 4$ gives 0.811 and 0.848; this control shows that the transplantation is no longer output-preserving at that depth (Table S3). The numerical check supports the exact cancellation argument; it is not a measured functional rank. Consequently, the E3 $L = 2$ stress sequence 1.0127, 1.4455, 2.0390 represents differently initialized and trained realizations of one output family, with different simulation cost, and is not a capacity trend (main Table 9). Section S8 gives the archived check and command.

The E1 nominally matched MLP counts exceed circuit output counts by 40.00% and 29.55%; spectral counts exceed them by 33.33% and 31.82% (Table S2). E1b reduces the MLP deviations to $-3.33\%$ and $+5.68\%$ (main Table 7). Its smaller spectral model is not matched, while the larger differs by $+10.23\%$. The implemented spectral search starts at width 2, giving 40 parameters; width 1 would give 21 but was not searched (main Tables 2 and 7).

Table S2: Circuit counts and E3 nearest nominal-count comparator designs. $\Delta_n = 100(P/P_{\text{nom}} - 1)$ and $\Delta_o = 100(P/P_{\text{out}} - 1)$ are classical-count deviations from the corresponding circuit counts. $P/P$ for classical models denotes the convention that no coordinates have been removed, not a Jacobian-rank estimate. MLP matches to targets 64 and 84 exceed the prespecified 10% tolerance.

| $(n_q, L)$ | VQC $P_{\text{nom}}/P_{\text{out}}$ | MLP $P/P$ | $\Delta_n$ (%) | $\Delta_o$ (%) | Spectral $P/P$ | $\Delta_n$ (%) | $\Delta_o$ (%) |
|---|---|---|---|---|---|---|---|
| 4, 2 | 44/30 | 42/42 | −4.55 | +40.00 | 40/40 | −9.09 | +33.33 |
| 4, 4 | 84/70 | 93/93 | +10.71 | +32.86 | 78/78 | −7.14 | +11.43 |
| 6, 2 | 64/30 | 57/57 | −10.94 | +90.00 | 59/59 | −7.81 | +96.67 |
| 6, 4 | 124/88 | 114/114 | −8.06 | +29.55 | 116/116 | −6.45 | +31.82 |
| 6, 8 | 244/208 | 249/249 | +2.05 | +19.71 | 249/249 | +2.05 | +19.71 |
| 8, 2 | 84/30 | 93/93 | +10.71 | +210.00 | 78/78 | −7.14 | +160.00 |

# S3 Hyperparameters, tuning, and resource accounting

Table S3: Detailed settings from the released code and declared execution environment. Launcher settings can be overridden and are not recorded within each JSON run. Main Table 3 gives the sampling distribution and loss weights.

| Item | Setting |
|---|---|
| Machine and arithmetic | Apple M2 Max, 12 CPU cores; `float64`/`complex128`; no accelerator |
| Software | Python 3.12.5; PyTorch 2.9.1; NumPy 2.5.1; SciPy 1.16.3 |
| E1 / rate selection | `OMP_NUM_THREADS=3`; E1 uses 3–4 concurrent processes |
| E1b launcher | `OMP_NUM_THREADS=2`, up to 3 processes |
| Cutoff launcher | Threads 2; reference serial, other jobs up to 4 |
| E3 main / auxiliary | Threads 3/2; main default workers 3; auxiliary v2 reads `logs/e3/aux_workers`, default 4 |
| E3 timing overlap | Author reports 7–11 concurrent processes for late circuit runs, load averages up to 161 on 12 cores; times are contention-affected observed durations, unsuitable for isolated-throughput comparisons |
| Dropped E3 jobs | VQC at $N_c = 4096$, seeds 0, 1; $4N_c = 16384$ fixed L-BFGS points; stopped after 2100 s, projected about 32400 s per seed; no completed result |
| Secondary-speed launcher | Threads 3, default workers 3 |
| E4 evaluation | Threads 4, author-declared execution setting; `run_e4.py` itself does not assign the thread variable |
| Adam | 2000 steps; cosine factor decays to 0.05; library-default momentum and stabilizer settings |
| L-BFGS | Step size 1; strong Wolfe; history 50; max iterations 1000; max evaluations 1250; gradient tolerance $10^{-12}$; change tolerance $10^{-14}$ |
| L-BFGS points | Fixed $4N_c$ collocation points and 512 shared boundary times; main E1/E1b $N_c = 512$ per Adam step |
| Rate selection | Five candidates, seed 100, Adam 1000 steps, no L-BFGS; fixed 2000 random validation points |
| Recorded history | Every 500 Adam steps, plus final L-BFGS record; `hist[-1].lbfgs_evals` counts closures when L-BFGS is used |

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| Item | Setting |
|---|---|
| Data-only sweep | 2000 Adam steps, no L-BFGS; fixed production training labels; no collocation or boundary loss |
| Data-only tuning labels | Rates selected once at $N_d = 256$ using 256 exact training samples and 2000 exact validation samples, shared by the five candidate rates; excluded from the production $N_d$ axis |
| Production validation | 2000 exact diagnostic samples per run; no update or checkpoint-selection role |
| Data-only unused precomputation | The shared training routine also constructs 4096 exact boundary samples; the data-only objective does not use them |
| Family-equivalence check | 64 random input pairs; transplant from 4 qubits to 6/8 qubits on indices $0, 1, 2, n_q - 1$; raw-output maxima $5.55 \times 10^{-16}/8.88 \times 10^{-16}$ at $L = 2$ and 0.811/0.848 at $L = 4$; executed `family_equivalence_check.txt` |

Table S4 reports every final candidate score, defined as the sum of the two relative validation errors. All selected values minimize their recorded candidate scores and agree with production rates. The full JSON retains intermediate Adam scores. Search-boundary selections limit any claim of optimal tuning. E3 physics and secondary-speed runs transfer the E1 exponential rates; data-only rates transfer from their separately selected label count. Thus E3 is a production training-label sweep conditional on separately selected hyperparameters. Its fixed tuning overhead is 256 exact training samples plus 2000 exact validation samples shared by the five candidate rates at $N_d = 256$ (Table S3). The tuning seed also makes these samples common to both model families. Production runs compute validation diagnostics, but those values neither drive parameter updates nor select checkpoints; the final state is saved. The plotted $N_d$ is therefore not a count of all exact-solution queries or total label-acquisition cost. In particular, the shared routine constructs unused exact boundary data also in the data-only regime (Table S3); the stated tuning overhead counts the exact samples used for training and rate selection.

Table S4: Complete final validation scores for the five learning-rate candidates. Each candidate uses seed 100 and 1000 Adam steps ($n = 1$ per candidate), so no replicate standard deviation is available. The selected rate minimizes the recorded validation score; all E1 and E1b selections agree with their executed rates. An asterisk marks a search-boundary choice. E3-data selects a data-only rate at $N_d = 256$, using 256 exact training samples and 2000 exact validation samples shared across the five rates; these tuning samples are outside the production training-label axis. The rows are tuning outcomes, not production-sweep outcomes.

| Set | Pulse | Config. | Model | $P_{\mathrm{nom}}/P_{\mathrm{out}}$ | 0.001 | 0.003 | 0.01 | 0.03 | 0.1 | Selected |
|---|---|---|---|---|---|---|---|---|---|---|
| E1 | exp | 4q2L | VQC | 44/30 | 3.4567 | 2.8118 | 2.5239 | 1.7015 | 1.4192 | 0.1* |
| E1 | exp | 4q2L | MLP | 42/42 | 2.3559 | 2.3412 | 2.4012 | 0.8587 | 0.7153 | 0.1* |
| E1 | exp | 4q2L | Spectral | 40/40 | 2.6825 | 2.5431 | 2.6223 | 1.9345 | 1.4690 | 0.1* |
| E1 | exp | 6q4L | VQC | 124/88 | 3.0510 | 2.2019 | 1.8525 | 1.3861 | 1.1540 | 0.1* |
| E1 | exp | 6q4L | MLP | 114/114 | 2.4258 | 2.2101 | 1.8108 | 0.9149 | 0.7805 | 0.1* |
| E1 | exp | 6q4L | Spectral | 116/116 | 2.6889 | 2.4889 | 1.4123 | 0.7417 | 0.4041 | 0.1* |
| E1 | exp | – | MLP-ref | 12802/12802 | 0.8554 | 0.4774 | 0.3556 | 0.5141 | 1.9452 | 0.01 |
| E1 | step | 4q2L | VQC | 44/30 | 3.2276 | 2.8055 | 2.6315 | 2.1116 | 2.2739 | 0.03 |
| E1 | step | 4q2L | MLP | 42/42 | 2.3828 | 2.3558 | 2.6608 | 1.2409 | 0.8261 | 0.1* |
| E1 | step | 4q2L | Spectral | 40/40 | 2.6899 | 2.8193 | 2.7902 | 2.8500 | 1.8826 | 0.1* |
| E1 | step | 6q4L | VQC | 124/88 | 2.9131 | 2.2049 | 2.0742 | 1.0767 | 1.1929 | 0.03 |
| E1 | step | 6q4L | MLP | 114/114 | 2.3629 | 2.2679 | 1.3758 | 0.9199 | 0.9586 | 0.03 |
| E1 | step | 6q4L | Spectral | 116/116 | 2.6969 | 2.4670 | 1.0896 | 0.7526 | 1.5595 | 0.03 |
| E1 | step | – | MLP-ref | 12802/12802 | 0.7504 | 0.4474 | 0.3991 | 0.4748 | 2.0660 | 0.01 |
| E1 | smooth | 4q2L | VQC | 44/30 | 2.4950 | 1.5693 | 1.4708 | 1.3955 | 1.3109 | 0.1* |
| E1 | smooth | 4q2L | MLP | 42/42 | 1.2797 | 1.5327 | 1.5691 | 1.5983 | 1.5815 | 0.001* |
| E1 | smooth | 4q2L | Spectral | 40/40 | 1.8194 | 1.5219 | 1.6730 | 1.6719 | 1.4684 | 0.1* |
| E1 | smooth | 6q4L | VQC | 124/88 | 2.2366 | 1.3259 | 1.1432 | 1.1599 | 1.2250 | 0.01 |
| E1 | smooth | 6q4L | MLP | 114/114 | 1.2549 | 1.3875 | 1.2753 | 1.0553 | 0.9920 | 0.1* |

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Table S4 – continued

| Set | Pulse | Config. | Model | $P_{\mathrm{nom}}/P_{\mathrm{out}}$ | 0.001 | 0.003 | 0.01 | 0.03 | 0.1 | Selected |
|---|---|---|---|---|---|---|---|---|---|---|
| E1 | smooth | 6q4L | Spectral | 116/116 | 1.5004 | 1.2704 | 1.0753 | 1.2484 | 1.4981 | 0.01 |
| E1 | smooth | – | MLP-ref | 12802/12802 | 1.1907 | 1.1183 | 0.8192 | 1.2301 | 1.7223 | 0.01 |
| E1b | exp | 4q2L | MLP | 29/29 | 2.4975 | 2.5621 | 2.1343 | 0.9171 | 0.8229 | 0.1* |
| E1b | exp | 4q2L | Spectral | 40/40 | 2.6825 | 2.5431 | 2.6223 | 1.9345 | 1.4690 | 0.1* |
| E1b | exp | 6q4L | MLP | 93/93 | 2.1514 | 1.7687 | 1.2053 | 0.7193 | 0.6650 | 0.1* |
| E1b | exp | 6q4L | Spectral | 97/97 | 2.6419 | 2.5871 | 1.2341 | 1.0738 | 0.7030 | 0.1* |
| E1b | step | 4q2L | MLP | 29/29 | 2.4389 | 2.6325 | 1.8126 | 1.0771 | 1.1280 | 0.03 |
| E1b | step | 4q2L | Spectral | 40/40 | 2.6899 | 2.8193 | 2.7902 | 2.8500 | 1.8826 | 0.1* |
| E1b | step | 6q4L | MLP | 93/93 | 2.1233 | 1.6029 | 1.3638 | 0.9004 | 0.9082 | 0.03 |
| E1b | step | 6q4L | Spectral | 97/97 | 2.7202 | 2.6010 | 1.6981 | 1.0486 | 0.6557 | 0.1* |
| E1b | smooth | 4q2L | MLP | 29/29 | 1.5330 | 1.4264 | 1.3949 | 1.5888 | 0.8332 | 0.1* |
| E1b | smooth | 4q2L | Spectral | 40/40 | 1.8194 | 1.5219 | 1.6730 | 1.6719 | 1.4684 | 0.1* |
| E1b | smooth | 6q4L | MLP | 93/93 | 1.2960 | 1.3951 | 1.6940 | 1.6959 | 0.9500 | 0.1* |
| E1b | smooth | 6q4L | Spectral | 97/97 | 1.5546 | 1.3645 | 1.1268 | 1.0184 | 0.9770 | 0.1* |
| E3-data | exp | 6q4L | VQC | 124/88 | 1.2484 | 0.8213 | 0.5597 | 0.3929 | 0.4193 | 0.03 |
| E3-data | exp | 6q4L | MLP | 114/114 | 1.1894 | 0.9734 | 0.4870 | 0.2921 | 0.1932 | 0.1* |

Table S5 distinguishes realized evaluations from iteration caps. The observed ranges are 1019–1250 for E1 and 385–1251 for E1b (records in Table S1); a configured evaluation limit therefore must not replace recorded line-search closure counts. The timer starts after model, reference-data, optimizer, and scheduler construction. It includes training, periodic and terminal validation, and final test-grid evaluation, ending before serialization and checkpoint writing. Shared-machine durations exclude selection and preparation cost.

Table S5: Realized L-BFGS closure calls and recorded runtime, mean ± sample standard deviation over the indicated number of trained seeds. Closure counts are `hist[-1].lbfgs_evals`; they count repeated objective evaluations rather than just optimizer iterations. Runtime covers training, validation, and final test-grid evaluation under concurrent shared-machine execution. It is not isolated-device or total-campaign cost.

| Set | Pulse | Config. | Model | $P_{\mathrm{nom}}/P_{\mathrm{out}}$ | $n$ | Closures | Runtime (s) |
|---|---|---|---|---|---|---|---|
| E1 | exp | 4q2L | VQC | 44/30 | 5 | $1161.0 \pm 79.8$ | $725.1 \pm 68.2$ |
| E1 | exp | 4q2L | MLP | 42/42 | 5 | $1085.8 \pm 16.4$ | $15.7 \pm 1.4$ |
| E1 | exp | 4q2L | Spectral | 40/40 | 5 | $1124.0 \pm 81.8$ | $45.0 \pm 3.2$ |
| E1 | exp | 6q4L | VQC | 124/88 | 5 | $1037.4 \pm 11.5$ | $4188.9 \pm 251.0$ |
| E1 | exp | 6q4L | MLP | 114/114 | 5 | $1046.4 \pm 10.5$ | $19.7 \pm 0.5$ |
| E1 | exp | 6q4L | Spectral | 116/116 | 5 | $1045.6 \pm 10.7$ | $44.1 \pm 3.7$ |
| E1 | exp | – | MLP-ref | 12802/12802 | 5 | $1062.6 \pm 5.7$ | $214.6 \pm 9.4$ |
| E1 | step | 4q2L | VQC | 44/30 | 5 | $1082.8 \pm 21.8$ | $624.5 \pm 15.5$ |
| E1 | step | 4q2L | MLP | 42/42 | 5 | $1073.0 \pm 10.1$ | $15.0 \pm 0.1$ |
| E1 | step | 4q2L | Spectral | 40/40 | 5 | $1071.2 \pm 18.1$ | $42.5 \pm 3.4$ |
| E1 | step | 6q4L | VQC | 124/88 | 5 | $1041.4 \pm 14.1$ | $3644.1 \pm 21.1$ |
| E1 | step | 6q4L | MLP | 114/114 | 5 | $1052.0 \pm 15.1$ | $21.2 \pm 2.8$ |
| E1 | step | 6q4L | Spectral | 116/116 | 5 | $1049.6 \pm 13.3$ | $47.3 \pm 3.4$ |
| E1 | step | – | MLP-ref | 12802/12802 | 5 | $1056.6 \pm 9.0$ | $212.6 \pm 7.8$ |
| E1 | smooth | 4q2L | VQC | 44/30 | 5 | $1158.2 \pm 80.8$ | $758.4 \pm 115.0$ |
| E1 | smooth | 4q2L | MLP | 42/42 | 5 | $1090.6 \pm 8.6$ | $14.6 \pm 1.0$ |
| E1 | smooth | 4q2L | Spectral | 40/40 | 5 | $1118.6 \pm 80.1$ | $43.1 \pm 8.7$ |
| E1 | smooth | 6q4L | VQC | 124/88 | 5 | $1030.8 \pm 9.5$ | $4473.7 \pm 466.5$ |
| E1 | smooth | 6q4L | MLP | 114/114 | 5 | $1044.4 \pm 14.1$ | $19.6 \pm 2.6$ |
| E1 | smooth | 6q4L | Spectral | 116/116 | 5 | $1040.2 \pm 12.8$ | $43.3 \pm 4.1$ |
| E1 | smooth | – | MLP-ref | 12802/12802 | 5 | $1057.2 \pm 10.3$ | $192.8 \pm 10.6$ |
| E1b | exp | 4q2L | MLP | 29/29 | 5 | $1169.6 \pm 55.4$ | $23.6 \pm 1.6$ |
| E1b | exp | 4q2L | Spectral | 40/40 | 5 | $1124.0 \pm 81.8$ | $65.5 \pm 14.7$ |
| E1b | exp | 6q4L | MLP | 93/93 | 5 | $1057.6 \pm 22.6$ | $18.9 \pm 0.5$ |
| E1b | exp | 6q4L | Spectral | 97/97 | 5 | $1065.8 \pm 15.6$ | $67.5 \pm 14.1$ |
| E1b | step | 4q2L | MLP | 29/29 | 5 | $1131.2 \pm 35.6$ | $13.0 \pm 0.5$ |
| E1b | step | 4q2L | Spectral | 40/40 | 5 | $1071.2 \pm 18.1$ | $44.0 \pm 3.9$ |
| E1b | step | 6q4L | MLP | 93/93 | 5 | $1053.6 \pm 8.0$ | $29.5 \pm 0.3$ |

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Table S5 – continued

| Set | Pulse | Config. | Model | $P_{\mathrm{nom}}/P_{\mathrm{out}}$ | $n$ | Closures | Runtime (s) |
|---|---|---|---|---|---|---|---|
| E1b | step | 6q4L | Spectral | 97/97 | 5 | $1057.4 \pm 10.4$ | $47.4 \pm 8.9$ |
| E1b | smooth | 4q2L | MLP | 29/29 | 5 | $982.2 \pm 334.8$ | $12.3 \pm 2.3$ |
| E1b | smooth | 4q2L | Spectral | 40/40 | 5 | $1118.6 \pm 80.1$ | $45.2 \pm 4.0$ |
| E1b | smooth | 6q4L | MLP | 93/93 | 5 | $1044.2 \pm 10.8$ | $18.5 \pm 0.2$ |
| E1b | smooth | 6q4L | Spectral | 97/97 | 5 | $1069.2 \pm 14.1$ | $49.0 \pm 5.4$ |

# S4 Additional field errors and initial-rate ablation

Table S6 retains maximum absolute errors and the minimum-stress-error seed in each production cell. Selection of a displayed field or E4 checkpoint does not replace all-seed reporting.

Table S6: Additional E1/E1b summaries. Maximum absolute field errors $a_T, a_\sigma$ are mean ± sample standard deviation across five seeds (0–4) on the common test grid. LR is the selected initial Adam rate used by all five runs. The last column gives the minimum relative stress error and its seed; these descriptive minima do not replace replicated means. Parameter counts list registered/output-affecting coordinates, not an independent functional dimension.

| Set | Pulse | Config. | Model | $P_{\mathrm{nom}}/P_{\mathrm{out}}$ | LR | $a_T$ | $a_\sigma$ | Best $e_\sigma$ (seed) |
|---|---|---|---|---|---|---|---|---|
| E1 | exp | 4q2L | VQC | 44/30 | 0.1 | $0.1248 \pm 0.0356$ | $1.0089 \pm 0.1748$ | 0.8460 (0) |
| E1 | exp | 4q2L | MLP | 42/42 | 0.1 | $0.0962 \pm 0.0501$ | $0.3323 \pm 0.0509$ | 0.2311 (1) |
| E1 | exp | 4q2L | Spectral | 40/40 | 0.1 | $0.1534 \pm 0.0250$ | $0.9340 \pm 0.2850$ | 0.3669 (3) |
| E1 | exp | 6q4L | VQC | 124/88 | 0.1 | $0.0559 \pm 0.0158$ | $0.7466 \pm 0.0486$ | 0.5698 (3) |
| E1 | exp | 6q4L | MLP | 114/114 | 0.1 | $0.0305 \pm 0.0208$ | $0.2265 \pm 0.0563$ | 0.1055 (2) |
| E1 | exp | 6q4L | Spectral | 116/116 | 0.1 | $0.0590 \pm 0.0329$ | $0.5314 \pm 0.1212$ | 0.3278 (2) |
| E1 | exp | – | MLP-ref | 12802/12802 | 0.01 | $0.0029 \pm 0.0006$ | $0.1407 \pm 0.0055$ | 0.0461 (0) |
| E1 | step | 4q2L | VQC | 44/30 | 0.03 | $0.2085 \pm 0.0726$ | $1.6738 \pm 0.3548$ | 0.8599 (1) |
| E1 | step | 4q2L | MLP | 42/42 | 0.1 | $0.1432 \pm 0.0146$ | $1.0282 \pm 0.0994$ | 0.4254 (3) |
| E1 | step | 4q2L | Spectral | 40/40 | 0.1 | $0.2836 \pm 0.0210$ | $1.5285 \pm 0.0853$ | 0.8714 (1) |
| E1 | step | 6q4L | VQC | 124/88 | 0.03 | $0.0815 \pm 0.0100$ | $1.4995 \pm 0.3731$ | 0.7391 (1) |
| E1 | step | 6q4L | MLP | 114/114 | 0.03 | $0.0890 \pm 0.0221$ | $0.7446 \pm 0.1917$ | 0.2300 (0) |
| E1 | step | 6q4L | Spectral | 116/116 | 0.03 | $0.1089 \pm 0.0179$ | $0.7461 \pm 0.0803$ | 0.3289 (2) |
| E1 | step | – | MLP-ref | 12802/12802 | 0.01 | $0.0120 \pm 0.0023$ | $0.1935 \pm 0.0124$ | 0.0439 (2) |
| E1 | smooth | 4q2L | VQC | 44/30 | 0.1 | $0.1016 \pm 0.0096$ | $0.8262 \pm 0.0671$ | 0.8986 (2) |
| E1 | smooth | 4q2L | MLP | 42/42 | 0.001 | $0.0675 \pm 0.0217$ | $0.4980 \pm 0.2913$ | 0.3627 (1) |
| E1 | smooth | 4q2L | Spectral | 40/40 | 0.1 | $0.1686 \pm 0.0409$ | $0.8171 \pm 0.0848$ | 0.7787 (3) |
| E1 | smooth | 6q4L | VQC | 124/88 | 0.01 | $0.0523 \pm 0.0070$ | $0.4936 \pm 0.0518$ | 0.4760 (4) |
| E1 | smooth | 6q4L | MLP | 114/114 | 0.1 | $0.0162 \pm 0.0051$ | $0.1613 \pm 0.0362$ | 0.1172 (3) |
| E1 | smooth | 6q4L | Spectral | 116/116 | 0.01 | $0.0647 \pm 0.0155$ | $0.5957 \pm 0.1561$ | 0.3706 (1) |
| E1 | smooth | – | MLP-ref | 12802/12802 | 0.01 | $0.0015 \pm 0.0004$ | $0.0421 \pm 0.0090$ | 0.0185 (3) |
| E1b | exp | 4q2L | MLP | 29/29 | 0.1 | $0.1004 \pm 0.0155$ | $0.4505 \pm 0.1842$ | 0.3135 (0) |
| E1b | exp | 4q2L | Spectral | 40/40 | 0.1 | $0.1534 \pm 0.0250$ | $0.9340 \pm 0.2850$ | 0.3669 (3) |
| E1b | exp | 6q4L | MLP | 93/93 | 0.1 | $0.0385 \pm 0.0045$ | $0.2561 \pm 0.0126$ | 0.2196 (0) |
| E1b | exp | 6q4L | Spectral | 97/97 | 0.1 | $0.0795 \pm 0.0289$ | $0.4972 \pm 0.0589$ | 0.6043 (2) |
| E1b | step | 4q2L | MLP | 29/29 | 0.03 | $0.1973 \pm 0.0736$ | $1.3764 \pm 0.4529$ | 0.4344 (4) |
| E1b | step | 4q2L | Spectral | 40/40 | 0.1 | $0.2836 \pm 0.0210$ | $1.5285 \pm 0.0853$ | 0.8714 (1) |
| E1b | step | 6q4L | MLP | 93/93 | 0.03 | $0.0755 \pm 0.0209$ | $0.4968 \pm 0.1402$ | 0.1644 (3) |
| E1b | step | 6q4L | Spectral | 97/97 | 0.1 | $0.1355 \pm 0.0224$ | $1.3076 \pm 0.3046$ | 0.3968 (1) |
| E1b | smooth | 4q2L | MLP | 29/29 | 0.1 | $0.0802 \pm 0.0176$ | $0.5411 \pm 0.0463$ | 0.4676 (4) |
| E1b | smooth | 4q2L | Spectral | 40/40 | 0.1 | $0.1686 \pm 0.0409$ | $0.8171 \pm 0.0848$ | 0.7787 (3) |
| E1b | smooth | 6q4L | MLP | 93/93 | 0.1 | $0.0323 \pm 0.0156$ | $0.3721 \pm 0.2498$ | 0.1427 (2) |
| E1b | smooth | 6q4L | Spectral | 97/97 | 0.1 | $0.0891 \pm 0.0229$ | $0.6996 \pm 0.1722$ | 0.5460 (3) |

Table S7 is the log-derived initial-rate-weight ablation. Both rate penalties vary together, and settings differ from E1. The main-paper cutoff table also changes both penalties. Neither ablation has replicated uncertainty. The cutoff comparison includes the circuit; the rate-weight ablation is confined to the classical models in Table S7.

Table S7: Initial-rate-weight ablation as reported in the execution log summary: exponential pulse, $\gamma = 9$, seed 0, $N_c = 1024$, learning rate 0.003. Each row is a single run ($n = 1$); no standard deviation is available. These log-derived summaries are not additional per-run entries in `runs_all.json`.

| Model | $P_{\mathrm{nom}}/P_{\mathrm{out}}$ | Rate weight | $e_T$ | $e_\sigma$ |
|---|---|---|---|---|
| MLP-ref | 12802/12802 | 0 | 0.0155 | 0.2152 |
| MLP-ref | 12802/12802 | 0.1 | 0.0169 | 0.0599 |
| MLP-ref | 12802/12802 | 1 | 0.0124 | 0.0492 |
| MLP | 114/114 | 0 | 0.1322 | 0.8132 |
| MLP | 114/114 | 0.1 | 0.0846 | 0.2688 |
| MLP | 114/114 | 1 | 0.1867 | 0.1827 |

## S5 Complete gradient-variance grid

Table S8 gives the retained variance estimates for the selected first-layer $R_Y$ derivative on qubit zero. Every objective uses the same fixed points and matched initialization seeds. Physics includes PDE, boundary, and initial-rate terms. The population denominator is used for this diagnostic, unlike the sample SD of replicated field errors. Individual gradient draws are unavailable; no variance-estimator confidence interval or slope-equivalence test is reported.

Table S8: E2 reported variance of the derivative with respect to the first-layer $R_Y$ angle on qubit 0, from 200 random initializations per cell, using denominator 200 (`ddof=0`) as established from the released code. The last column is the ordinary least-squares slope of $\log_{10}(\mathrm{Var})$ versus qubit count over the displayed range. These are variance estimates, not mean ± standard deviation summaries; individual gradient draws and uncertainty intervals are not retained in the aggregate records.

| $L$ | Loss | $n_q = 2$ | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 2 | data | $9.56 \times 10^{-1}$ | $7.54 \times 10^{-1}$ | $5.05 \times 10^{-1}$ | $7.07 \times 10^{-1}$ | $6.00 \times 10^{-1}$ |
| 2 | physics | $5.89 \times 10^{0}$ | $9.31 \times 10^{0}$ | $4.66 \times 10^{0}$ | $7.47 \times 10^{0}$ | $1.05 \times 10^{1}$ |
| 2 | combined | $9.01 \times 10^{0}$ | $1.25 \times 10^{1}$ | $6.54 \times 10^{0}$ | $1.03 \times 10^{1}$ | $1.36 \times 10^{1}$ |
| 4 | data | $2.94 \times 10^{-1}$ | $1.00 \times 10^{-1}$ | $6.45 \times 10^{-2}$ | $6.59 \times 10^{-2}$ | $7.77 \times 10^{-2}$ |
| 4 | physics | $4.87 \times 10^{1}$ | $2.79 \times 10^{1}$ | $9.30 \times 10^{0}$ | $2.48 \times 10^{1}$ | $7.30 \times 10^{0}$ |
| 4 | combined | $5.05 \times 10^{1}$ | $2.89 \times 10^{1}$ | $9.68 \times 10^{0}$ | $2.53 \times 10^{1}$ | $7.80 \times 10^{0}$ |
| 8 | data | $1.41 \times 10^{-1}$ | $3.26 \times 10^{-2}$ | $1.50 \times 10^{-2}$ | $5.67 \times 10^{-3}$ | $3.84 \times 10^{-3}$ |
| 8 | physics | $3.83 \times 10^{2}$ | $2.76 \times 10^{2}$ | $1.25 \times 10^{2}$ | $5.28 \times 10^{1}$ | $2.13 \times 10^{1}$ |
| 8 | combined | $3.85 \times 10^{2}$ | $2.78 \times 10^{2}$ | $1.26 \times 10^{2}$ | $5.30 \times 10^{1}$ | $2.14 \times 10^{1}$ |

| $L$ | Loss | $n_q = 7$ | 8 | 9 | 10 | Slope |
|---|---|---|---|---|---|---|
| 2 | data | $7.83 \times 10^{-1}$ | $6.61 \times 10^{-1}$ | $6.21 \times 10^{-1}$ | $7.22 \times 10^{-1}$ | −0.008 |
| 2 | physics | $8.65 \times 10^{0}$ | $3.43 \times 10^{0}$ | $1.10 \times 10^{1}$ | $1.47 \times 10^{1}$ | 0.027 |
| 2 | combined | $1.14 \times 10^{1}$ | $5.88 \times 10^{0}$ | $1.37 \times 10^{1}$ | $1.72 \times 10^{1}$ | 0.020 |
| 4 | data | $5.62 \times 10^{-2}$ | $7.28 \times 10^{-2}$ | $7.19 \times 10^{-2}$ | $9.93 \times 10^{-2}$ | −0.038 |
| 4 | physics | $1.65 \times 10^{1}$ | $1.08 \times 10^{1}$ | $1.17 \times 10^{1}$ | $7.04 \times 10^{0}$ | −0.076 |
| 4 | combined | $1.70 \times 10^{1}$ | $1.15 \times 10^{1}$ | $1.23 \times 10^{1}$ | $7.62 \times 10^{0}$ | −0.074 |
| 8 | data | $2.48 \times 10^{-3}$ | $2.20 \times 10^{-3}$ | $2.30 \times 10^{-3}$ | $2.73 \times 10^{-3}$ | −0.206 |
| 8 | physics | $8.72 \times 10^{0}$ | $1.02 \times 10^{1}$ | $8.99 \times 10^{0}$ | $1.06 \times 10^{1}$ | −0.227 |
| 8 | combined | $8.80 \times 10^{0}$ | $1.03 \times 10^{1}$ | $9.09 \times 10^{0}$ | $1.07 \times 10^{1}$ | −0.227 |

The slopes in the main paper are ordinary least-squares fits to the full-precision variances, not the rounded cells above. The additional deepest-circuit fit over the largest qubit counts is reported

Table S9: Minimum and maximum physics/data ratios of the E2 reported gradient variances across the tested qubit counts. These ratios use the full-precision reported variances and depend on the stated loss normalization.

| $L$ | Minimum variance ratio | Maximum variance ratio |
|---|---|---|
| 2 | 5.19 | 20.31 |
| 4 | 70.89 | 375.41 |
| 8 | 2726.20 | 9311.63 |

in main Table 8. Loss scaling affects variance magnitude quadratically, and no asymptotic exponent is identified from these finite scans.

## S6 Selected checkpoints and complete noise results

All tables in this section use $\gamma = 9$, the $41 \times 40$ noise grid, and the same selected checkpoint pairs. The exact controls are actual $p = 0$, `shots="exact"` evaluations. Stored E1-grid clean metadata are not substituted. The full records are grouped by wave speed as well as pulse and configuration. A shot count is a per-input, per-observable measurement budget; the two readouts are sampled independently in this evaluator. Sample SDs describe the two selected checkpoints and do not estimate repeated-shot uncertainty.

Table S10: Selected E4 checkpoint identities, same-grid clean errors, and affine-readout coefficients. Each row is one trained checkpoint; clean errors use $p = 0$ and exact expectations on the E4 grid. Readout indices $0, 1$ denote temperature and stress. Gains and offsets are the supplied checkpoint values, rounded to the displayed precision. $P_{\text{nom}}/P_{\text{out}}$ does not assert an independent functional dimension.

| Pulse | Config. | $P_{\text{nom}}/P_{\text{out}}$ | Seed | $e_T$ | $e_\sigma$ | $\alpha_0$ | $\alpha_1$ | $\beta_0$ | $\beta_1$ |
|---|---|---|---|---|---|---|---|---|---|
| exp | 4q2L | 44/30 | 0 | 0.2624 | 0.8576 | 43.90 | 37.31 | 0.015 | 0.046 |
| exp | 4q2L | 44/30 | 3 | 0.3993 | 1.0458 | 198.51 | 207.61 | 0.000 | 0.057 |
| exp | 6q4L | 124/88 | 1 | 0.1376 | 0.5921 | 3.55 | 15.60 | 0.003 | −0.207 |
| exp | 6q4L | 124/88 | 3 | 0.1814 | 0.5686 | 3.80 | 12.67 | 0.027 | −0.254 |
| step | 4q2L | 44/30 | 1 | 0.7394 | 0.8597 | 50.73 | 113.36 | 0.138 | −0.184 |
| step | 4q2L | 44/30 | 2 | 0.4912 | 0.8911 | 3.96 | 20.59 | 0.099 | −3.246 |
| step | 6q4L | 124/88 | 0 | 0.2333 | 0.8246 | 1.78 | 5.76 | −0.027 | −0.444 |
| step | 6q4L | 124/88 | 1 | 0.2189 | 0.7432 | 2.20 | 11.89 | 0.037 | 0.854 |
| smooth | 4q2L | 44/30 | 2 | 0.3383 | 0.8935 | −236.17 | 82.77 | −4.051 | 0.025 |
| smooth | 4q2L | 44/30 | 4 | 0.3115 | 0.9153 | −1.96 | 0.95 | 0.050 | −0.135 |
| smooth | 6q4L | 124/88 | 1 | 0.1703 | 0.5325 | 2.95 | 6.45 | −0.007 | 0.138 |
| smooth | 6q4L | 124/88 | 4 | 0.1231 | 0.4730 | 1.29 | 6.53 | 0.003 | −0.261 |

Table S11: Corrected E4 inference errors on the common $41 \times 40$ grid, including the zero-noise exact-expectation controls, finite shots alone, and depolarization with exact expectations. Entries are mean ± sample standard deviation across the same two selected trained checkpoints per pulse/configuration. Exact denotes expectation evaluation. Each $p = 0$, exact row is the clean control for its block; the stored `clean_T` and `clean_sigma` metadata are not used as these controls.

| Pulse | Config. | $p$ | Shots | $e_T$ | $e_\sigma$ |
|---|---|---|---|---|---|
| exp | 4q2L | 0 | exact | $0.3308 \pm 0.0968$ | $0.9517 \pm 0.1331$ |
| exp | 4q2L | 0 | 1000 | $73.8240 \pm 67.6805$ | $21.3149 \pm 20.9250$ |

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Table S11 – continued

| Pulse | Config. | $p$ | Shots | $e_T$ | $e_\sigma$ |
|---|---|---|---|---|---|
| exp | 4q2L | 0 | 10000 | $22.4574 \pm 20.3642$ | $6.9947 \pm 6.7734$ |
| exp | 4q2L | 0 | 100000 | $7.1757 \pm 6.3994$ | $2.4748 \pm 1.9383$ |
| exp | 4q2L | 0.001 | exact | $0.3290 \pm 0.0966$ | $0.9500 \pm 0.1333$ |
| exp | 4q2L | 0.01 | exact | $0.3133 \pm 0.0949$ | $0.9346 \pm 0.1350$ |
| exp | 6q4L | 0 | exact | $0.1595 \pm 0.0309$ | $0.5804 \pm 0.0166$ |
| exp | 6q4L | 0 | 1000 | $2.2119 \pm 0.0068$ | $2.5758 \pm 0.2728$ |
| exp | 6q4L | 0 | 10000 | $0.7053 \pm 0.0582$ | $0.9735 \pm 0.0950$ |
| exp | 6q4L | 0 | 100000 | $0.2691 \pm 0.0321$ | $0.6299 \pm 0.0251$ |
| exp | 6q4L | 0.001 | exact | $0.1556 \pm 0.0282$ | $0.5753 \pm 0.0184$ |
| exp | 6q4L | 0.01 | exact | $0.1588 \pm 0.0336$ | $0.6135 \pm 0.1210$ |
| step | 4q2L | 0 | exact | $0.6153 \pm 0.1755$ | $0.8754 \pm 0.0222$ |
| step | 4q2L | 0 | 1000 | $14.8745 \pm 17.8301$ | $7.5606 \pm 7.1462$ |
| step | 4q2L | 0 | 10000 | $4.7963 \pm 5.5767$ | $2.5643 \pm 2.0159$ |
| step | 4q2L | 0 | 100000 | $1.6668 \pm 1.5980$ | $1.2036 \pm 0.3992$ |
| step | 4q2L | 0.001 | exact | $0.6134 \pm 0.1769$ | $0.8710 \pm 0.0171$ |
| step | 4q2L | 0.01 | exact | $0.5974 \pm 0.1906$ | $0.8566 \pm 0.0056$ |
| step | 6q4L | 0 | exact | $0.2261 \pm 0.0102$ | $0.7839 \pm 0.0576$ |
| step | 6q4L | 0 | 1000 | $1.0864 \pm 0.1215$ | $1.2875 \pm 0.3769$ |
| step | 6q4L | 0 | 10000 | $0.3964 \pm 0.0344$ | $0.8455 \pm 0.0039$ |
| step | 6q4L | 0 | 100000 | $0.2514 \pm 0.0043$ | $0.7912 \pm 0.0535$ |
| step | 6q4L | 0.001 | exact | $0.2214 \pm 0.0053$ | $0.7749 \pm 0.0519$ |
| step | 6q4L | 0.01 | exact | $0.2029 \pm 0.0140$ | $0.7339 \pm 0.0261$ |
| smooth | 4q2L | 0 | exact | $0.3249 \pm 0.0189$ | $0.9044 \pm 0.0154$ |
| smooth | 4q2L | 0 | 1000 | $73.0238 \pm 101.5526$ | $8.6281 \pm 10.8824$ |
| smooth | 4q2L | 0 | 10000 | $23.3152 \pm 32.2638$ | $2.9909 \pm 2.9338$ |
| smooth | 4q2L | 0 | 100000 | $7.5194 \pm 10.1558$ | $1.3653 \pm 0.6361$ |
| smooth | 4q2L | 0.001 | exact | $0.3898 \pm 0.1111$ | $0.9046 \pm 0.0157$ |
| smooth | 4q2L | 0.01 | exact | $1.2674 \pm 1.3552$ | $0.9069 \pm 0.0185$ |
| smooth | 6q4L | 0 | exact | $0.1467 \pm 0.0334$ | $0.5027 \pm 0.0420$ |
| smooth | 6q4L | 0 | 1000 | $1.2773 \pm 0.7026$ | $1.3724 \pm 0.0278$ |
| smooth | 6q4L | 0 | 10000 | $0.4320 \pm 0.2159$ | $0.6369 \pm 0.0212$ |
| smooth | 6q4L | 0 | 100000 | $0.1949 \pm 0.0692$ | $0.5163 \pm 0.0425$ |
| smooth | 6q4L | 0.001 | exact | $0.1539 \pm 0.0310$ | $0.5041 \pm 0.0386$ |
| smooth | 6q4L | 0.01 | exact | $0.2357 \pm 0.0060$ | $0.5319 \pm 0.0026$ |

Table S12: Corrected E4 joint shot and depolarizing-noise errors on the same evaluation grid and selected checkpoints as Table S11. Entries are mean ± sample standard deviation across two trained seeds. The dispersion does not estimate uncertainty over repeated independent shot-sampling realizations.

| Pulse | Config. | $p$ | Shots | $e_T$ | $e_\sigma$ |
|---|---|---|---|---|---|
| exp | 4q2L | 0.001 | 1000 | $73.9079 \pm 68.5329$ | $21.3403 \pm 20.8135$ |
| exp | 4q2L | 0.001 | 10000 | $22.3959 \pm 20.3455$ | $7.0580 \pm 6.8084$ |
| exp | 4q2L | 0.001 | 100000 | $7.2117 \pm 6.4485$ | $2.4538 \pm 1.9118$ |
| exp | 4q2L | 0.01 | 1000 | $73.8193 \pm 68.6636$ | $21.3438 \pm 20.8092$ |
| exp | 4q2L | 0.01 | 10000 | $22.5016 \pm 20.4681$ | $6.9110 \pm 6.7321$ |
| exp | 4q2L | 0.01 | 100000 | $7.1371 \pm 6.4137$ | $2.4750 \pm 1.9450$ |
| exp | 6q4L | 0.001 | 1000 | $2.1775 \pm 0.1358$ | $2.6298 \pm 0.3841$ |
| exp | 6q4L | 0.001 | 10000 | $0.7064 \pm 0.0501$ | $0.9714 \pm 0.0901$ |
| exp | 6q4L | 0.001 | 100000 | $0.2721 \pm 0.0251$ | $0.6185 \pm 0.0168$ |
| exp | 6q4L | 0.01 | 1000 | $2.2078 \pm 0.0967$ | $2.5795 \pm 0.3999$ |

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Table S12 – continued

| Pulse | Config. | $p$ | Shots | $e_T$ | $e_\sigma$ |
|---|---|---|---|---|---|
| exp | 6q4L | 0.01 | 10000 | $0.7039 \pm 0.0430$ | $0.9982 \pm 0.1689$ |
| exp | 6q4L | 0.01 | 100000 | $0.2697 \pm 0.0008$ | $0.6630 \pm 0.1217$ |
| step | 4q2L | 0.001 | 1000 | $14.8401 \pm 17.8334$ | $7.5581 \pm 7.1904$ |
| step | 4q2L | 0.001 | 10000 | $4.7419 \pm 5.4846$ | $2.5953 \pm 2.0907$ |
| step | 4q2L | 0.001 | 100000 | $1.6123 \pm 1.5334$ | $1.2206 \pm 0.4446$ |
| step | 4q2L | 0.01 | 1000 | $14.7380 \pm 17.7327$ | $7.5403 \pm 7.1826$ |
| step | 4q2L | 0.01 | 10000 | $4.7569 \pm 5.5249$ | $2.5909 \pm 2.0626$ |
| step | 4q2L | 0.01 | 100000 | $1.6660 \pm 1.6301$ | $1.1910 \pm 0.4320$ |
| step | 6q4L | 0.001 | 1000 | $1.0987 \pm 0.1280$ | $1.2799 \pm 0.3607$ |
| step | 6q4L | 0.001 | 10000 | $0.3963 \pm 0.0399$ | $0.8458 \pm 0.0211$ |
| step | 6q4L | 0.001 | 100000 | $0.2488 \pm 0.0017$ | $0.7817 \pm 0.0465$ |
| step | 6q4L | 0.01 | 1000 | $1.0939 \pm 0.2022$ | $1.2507 \pm 0.3792$ |
| step | 6q4L | 0.01 | 10000 | $0.3936 \pm 0.0455$ | $0.8039 \pm 0.0761$ |
| step | 6q4L | 0.01 | 100000 | $0.2300 \pm 0.0181$ | $0.7407 \pm 0.0318$ |
| smooth | 4q2L | 0.001 | 1000 | $73.0445 \pm 101.5041$ | $8.6282 \pm 10.8859$ |
| smooth | 4q2L | 0.001 | 10000 | $23.9366 \pm 33.1582$ | $3.0050 \pm 2.9535$ |
| smooth | 4q2L | 0.001 | 100000 | $7.3960 \pm 9.9820$ | $1.3765 \pm 0.6522$ |
| smooth | 4q2L | 0.01 | 1000 | $74.1424 \pm 103.1490$ | $8.4736 \pm 10.6587$ |
| smooth | 4q2L | 0.01 | 10000 | $23.3594 \pm 32.3435$ | $3.0202 \pm 2.9673$ |
| smooth | 4q2L | 0.01 | 100000 | $7.4067 \pm 9.9998$ | $1.4025 \pm 0.6822$ |
| smooth | 6q4L | 0.001 | 1000 | $1.2904 \pm 0.7148$ | $1.3423 \pm 0.0080$ |
| smooth | 6q4L | 0.001 | 10000 | $0.4415 \pm 0.2206$ | $0.6299 \pm 0.0275$ |
| smooth | 6q4L | 0.001 | 100000 | $0.2015 \pm 0.0657$ | $0.5175 \pm 0.0384$ |
| smooth | 6q4L | 0.01 | 1000 | $1.3081 \pm 0.7323$ | $1.3870 \pm 0.0219$ |
| smooth | 6q4L | 0.01 | 10000 | $0.4791 \pm 0.1837$ | $0.6600 \pm 0.0052$ |
| smooth | 6q4L | 0.01 | 100000 | $0.2708 \pm 0.0312$ | $0.5473 \pm 0.0029$ |

# S7 Corrected-solution verification

The main appendix gives the corrected reference formulas. The scripts `closed_form_corrected.py`, `laplace_ref.py`, `fd_ref.py`, and `check_corrected.py` supply algebraically independent inverse evaluation and a separately discretized evolution calculation. The pointwise log is `verification_log_closed_form.txt`. Tables below report what was executed; they do not imply a new experiment or a uniform error bound. Numerical inversion uses the de Hoog method (de Hoog et al., 1982).

## S7.1 Reference-solution corrections and executed verification

Table S13 records the correction inventory supplied by the author for Al-Qahtani (2026). The corrected expressions are given in the main-text exact-solution appendix; the inventory is not a new inspection of the original published typesetting. Tables S14 and S15 summarize the executed driver and its retained log. The complete pointwise output is supplied as `verification_log_closed_form.txt`.

The supplied finite-difference check evolves $w_{tt} = \gamma(w - \gamma T)_{xx}$ and recovers $\sigma = w - \gamma T$. The checks in Table S15 support agreement at the sampled points, but do not provide a uniform error bound or a mesh-convergence study. No unrounded maximum inversion discrepancy is retained for the exponential or rectangular pulse. The large-$\gamma$ block checks finiteness only and does not replace the secondary-wave-speed network experiments.

Table S13: Correction inventory from the supplied author audit. Equation and figure identifiers in the first column belong to Al-Qahtani (2026). The energy values describe that paper's caption; they are not the pulse energies used for the present benchmark.

| Original location | Correction or convention used here |
|---|---|
| Equation (11) | Use the complementary-error-function argument $(2t-x)/(2\sqrt{t})$ in the identified term; the reported opposite sign violates the zero initial-temperature condition. |
| Equation (12) | Replace the reported unit-step response, whose initial limit is $e^{-x}$, by the corrected unit-step response $\mathcal{T}_0$. |
| Equations (19)–(26) | Replace the reported stress expressions containing uncanceled growing contributions by the corrected stress response in the main-text exact-solution appendix. |
| Equation (15) | With the original stress scale $\Sigma_0 = b_{\mathrm{th}} I_1/(k\delta)$, the original dimensionless stress $s$ satisfies $s_{xx} - s_{tt}/\gamma = T_{tt}/\gamma$. The present convention $\sigma = \gamma s$ gives unit forcing coefficient. |
| Figure 2 caption | The reported products are $A_s\Delta = 0.25$ and $A_e/\beta = 0.5$, so the two reported energies are unequal. |

Table S14: Executed reference-check settings, recovered from `check_corrected.py`, `laplace_ref.py`, `fd_ref.py`, and the verification log. All space–time quantities are dimensionless. The inverse transform is evaluated with the stated arithmetic precision and converted to double precision before comparison.

| Check or setting | Recorded specification |
|---|---|
| Exponential and rectangular checks | $\gamma = 9$; exponential $\beta = 2$, $A_e = 1$; rectangular $A_s = 1$, $\Delta = 0.5$. |
| Sampling for those checks | $x \in \{0.1, 0.5, 1.1, 2.1, 4.1\}$ and $t \in \{0.1, 0.3, 0.75, 1.5, 2.5\}$; Cartesian product, 25 locations per pulse. |
| Smooth inversion check | $\gamma = 9$, $A = 4/3$, $\beta_1 = 2$, $\beta_2 = 8$; $(x,t) \in \{(0.1,0.1),(1.1,0.75),(2.1,1.5),(0.5,2.5),(4.1,2.5)\}$; 5 locations. |
| Inverse-transform procedure | de Hoog method, 30-decimal-digit arithmetic; exponential/rectangular values printed with 7 significant digits; smooth values with 8. The smooth field transforms use the pulse transform $\widehat{f}(s) = (4/3)[(s+2)^{-1} - (s+8)^{-1}]$. |
| Finite-difference discretization | $x \in [0, 24]$, 2400 spatial intervals, $\Delta x = 0.01$; centered interior differences and classical fourth-order Runge–Kutta; $\Delta t \simeq 0.2(\Delta x)^2 = 2 \times 10^{-5}$; final time 2.5. |
| Finite-difference fields and data | Evolve $T, w, w_t$ with $w = \sigma + \gamma T$; $T(x,0) = w(x,0) = w_t(x,0) = 0$; $T_x(0,t) = 0$, $w(0,t) = \gamma T(0,t)$; $T(24,t) = w(24,t) = 0$. |
| Large-$\gamma$ finite-value check | Exponential pulse, $\beta = 2$, $\gamma = 100$; $x \in \{0, 0.1, 1, 5, 20\}$ and $t \in \{0.05, 0.5, 2, 5\}$; Cartesian product, 20 stress evaluations, without an independent reference. |

Table S15: Results retained in the verification log. Finite-difference maxima are over the sampled locations in Table S14, not the full discretization grid. The smooth maximum is the maximum absolute discrepancy across both fields and all its sampled locations. These are deterministic reference checks and are not training errors.

| Method/case | Temperature result | Stress result |
|---|---|---|
| de Hoog, exponential | Equal at 7 printed significant digits at all 25 locations | Equal at 7 printed significant digits at all 25 locations |
| de Hoog, rectangular | Equal at 7 printed significant digits at all 25 locations | Equal at 7 printed significant digits at all 25 locations |
| Finite differences, exponential | Maximum absolute discrepancy $4.20 \times 10^{-6}$ | Maximum absolute discrepancy $6.18 \times 10^{-3}$ |
| Finite differences, rectangular | Maximum absolute discrepancy $5.91 \times 10^{-6}$ | Maximum absolute discrepancy $6.19 \times 10^{-3}$ |
| de Hoog, smooth | Maximum absolute discrepancy $5.13 \times 10^{-16}$ over both fields at the 5 locations; no smooth finite-difference comparison is logged. | |
| Large-$\gamma$ check | Not evaluated in this check | All 20 reported stress values are finite; no accuracy bound is established. |
| Module regression self-test | `SELF-TEST PASSED`; maximum absolute discrepancy $4.0 \times 10^{-10}$ over 5 stored reference points, below the $10^{-9}$ threshold. Both fields are finite for all pulses on the $241 \times 240$ grid. Stored constants are rounded inversion references. | |

The appended `ground_truth.py` self-test was executed in the declared Python 3.12.5, NumPy 2.5.1, and SciPy 1.16.3 environment. At its five stored point references, the largest deviation among exponential/rectangular temperatures and stresses is $4.0 \times 10^{-10}$, below its $10^{-9}$ pass threshold (Table S15). These references are stored as rounded constants derived from the inverse-Laplace calculation; this replay is a regression check of the module, not an additional independent reference method. It also verifies finite values of both fields for all pulses on the $241 \times 240$ grid. The execution stdout is retained in the archive verification log.

## S8 Run commands and reconstruction

The entry points below are provided in the analysis directory of `QPINN_LTE_release_v1_2026-10-21.zip`. The archive contains their results, checkpoint state dictionaries, execution logs, and the package listed in Section S1. Its `RELEASE_README` supplies the table/figure to input/data/generator mapping and regeneration commands; the analysis version used here is described by `results_package_v6.md`. These commands document the executed workflow and are run from the assembled analysis directory with its dependencies. The launchers retain the thread settings in Table S3.

```
python3 check_corrected.py
python3 ground_truth.py
python3 functional_count.py
OMP_NUM_THREADS=2 python3 family_equivalence.py
mkdir -p logs/lrsel logs/e1
```

```
zsh run_lrsel.sh
zsh run_lrsel_hi.sh
python3 select_lrs.py
zsh run_e1.sh
zsh run_functional_matched.sh
zsh run_ablation_xmin.sh mlpref
zsh run_ablation_xmin.sh circuit
python3 qpinn_local.py e2 --case exp --gamma 9
OMP_NUM_THREADS=4 python3 run_e4.py
zsh run_gamma.sh
unset SKIP_PENDING
python3 make_results_package.py
python3 make_runs_all.py
```

`family_equivalence_check.txt` contains the executed output of `family_equivalence.py`, discussed in Section S2. The script compares the two raw circuit outputs after active-qubit parameter transplantation; it does not train any model. The code and retained output accompany the coordinate audit and reference-solution verification records.

The E3 source launcher `run_e3.sh` performs its data-only rate selection and generates the planned job list in `logs/e3/jobs.txt`. The historical main and replacement auxiliary entry points were `zsh run_e3.sh` and `zsh run_e3_aux.sh`, respectively. Its unmodified list still contains the dropped high-collocation circuit jobs (Table S3); a fresh execution would therefore launch the full plan. To replay the retained experiment, omit those two job lines before dispatch or reconstruct the retained jobs from the per-run settings in Table S16. The auxiliary `run_e3_aux.sh` v2 orders remaining jobs by its cost heuristic, reads its adjustable worker count from `logs/e3/aux_workers`, and skips any job whose log already exists. Its process pattern counts E3 processes, not only circuits despite that wording in a comment. An existing log or final queue marker is not a completion certificate; retained cells were checked against their JSON records and declared seeds (Table S1).

The reference reproducibility check used the following settings; the command generates the separately retained replay record rather than replacing its cutoff-ablation counterpart (main Table 4).

```
OMP_NUM_THREADS=3 python3 qpinn_local.py run --model mlpref \
  --case exp --gamma 9 --n_q 6 --L 4 --seed 0 \
  --iters 2000 --lbfgs 1000 --n_c 512 --lr 0.01 \
  --ic_xmin 0.3 --tag xmin0.3_omp3 --out results/ablation_repro
```

To reconstruct summaries, group production rows by experiment, pulse, wave speed, model, architecture, regime, and swept setting. Preserve the declared seeds. Use arithmetic means and sample SDs for E1/E1b and secondary speeds, means and both seed values or ranges for E3, and individual values for ablations. Metrics are `relL2_T`, `relL2_sigma`, `maxabs_T`, `maxabs_sigma`, and `train_s`; realized L-BFGS closures are `hist[-1].lbfgs_evals` when present. Data-only runs have no L-BFGS stage. Best seeds minimize `relL2_sigma`. Selection groups use minimum final `val`; E2 fits use unrounded `var` values. E4 groups additionally include `p` and `shots`, retaining clean exact rows. The reported E4 set is restricted to $\gamma = 9$ (Table S1); rerunning `run_e4.py` after secondary checkpoints exist can evaluate additional speeds, whose records must be kept separate. Candidate runs and superseded outputs remain separate from production summaries. A parser's unused default `n_c` in a data-only record is not actual collocation. The generators tolerate partial groups, so the inventory and explicit per-seed audit, rather than generator success alone, establish completion.

# S9 Efficiency and secondary-speed per-run records

## S9.1 E3: all executed runs

Table S16 contains all 66 E3 records once. Design codes $P_{q,L}$ denote parameter sweeps: $N_c = 512$, $4N_c = 2048$, and rate 0.1. Codes $C_j$ denote the $(6, 4)$ collocation design with $N_c = j$, $4N_c = 4j$, and rate 0.1. Codes $D_j$ denote the $(6, 4)$ data design with $N_d = j$ production training labels, no collocation or L-BFGS, and rates 0.03 (VQC) and 0.1 (MLP). These rates were selected separately using 256 exact training samples and 2000 exact validation samples shared across five candidate rates (Section S3); those samples are not included in $N_d$. Production validation is diagnostic and neither updates parameters nor selects checkpoints. All use 2000 Adam steps. The collocation baseline $C_{512}$ is $P_{6,4}$, not another run. Parameter counts and both matching deviations are given in Table S2; all $C_j, D_j$ rows use its $(6, 4)$ counts. "Index" is the zero-based record index in the released `runs_all.json`. It identifies each full record and its stored `name` and `file`. The 84-target classical executions have identical errors and histories but distinct measured times; all executions are shown for provenance and are not independent designs. Times are contention-affected observed durations, unsuitable for isolated-throughput comparisons; overlapping queues affect these wall-clock observations.

Table S16: E3 per-seed outcomes, timing, and actual L-BFGS closure counts. A zero closure count denotes the data-only regime. No sample standard deviations are assigned to the two-seed cells. Timings are contention-affected observed durations, unsuitable for isolated-throughput comparisons. Data-design $N_d$ counts production training labels, conditional on the separate tuning overhead specified in Section S3.

| Index | Design | Model | Seed | $e_T$ | $e_\sigma$ | Time (s) | Closures |
|---|---|---|---|---|---|---|---|
| 223 | $P_{4,2}$ | VQC | 0 | 0.2561 | 0.8460 | 943.07 | 1250 |
| 224 | $P_{4,2}$ | VQC | 1 | 0.5500 | 1.1794 | 730.79 | 1159 |
| 195 | $P_{4,2}$ | MLP | 0 | 0.2723 | 0.3823 | 13.06 | 1066 |
| 196 | $P_{4,2}$ | MLP | 1 | 0.2788 | 0.2311 | 13.33 | 1079 |
| 175 | $P_{4,2}$ | Spectral | 0 | 0.4112 | 0.9487 | 33.71 | 1244 |
| 176 | $P_{4,2}$ | Spectral | 1 | 0.4001 | 0.7746 | 31.57 | 1172 |
| 225 | $P_{4,4}$ | VQC | 0 | 0.1905 | 1.0473 | 1437.31 | 1039 |
| 226 | $P_{4,4}$ | VQC | 1 | 0.1990 | 0.9030 | 1328.29 | 1019 |
| 197 | $P_{4,4}$ | MLP | 0 | 0.1100 | 0.2196 | 16.49 | 1046 |
| 198 | $P_{4,4}$ | MLP | 1 | 0.1005 | 0.2610 | 16.57 | 1044 |
| 177 | $P_{4,4}$ | Spectral | 0 | 0.3864 | 0.6899 | 32.07 | 1057 |
| 178 | $P_{4,4}$ | Spectral | 1 | 0.1796 | 0.4204 | 29.97 | 1037 |
| 227 | $P_{6,2}$ | VQC | 0 | 0.6498 | 1.3497 | 2352.39 | 1099 |
| 228 | $P_{6,2}$ | VQC | 1 | 0.6162 | 1.5413 | 3342.69 | 1250 |
| 199 | $P_{6,2}$ | MLP | 0 | 0.1020 | 0.2854 | 15.14 | 1057 |
| 200 | $P_{6,2}$ | MLP | 1 | 0.2040 | 0.4341 | 14.01 | 1055 |
| 179 | $P_{6,2}$ | Spectral | 0 | 0.4830 | 0.8510 | 31.42 | 1064 |
| 180 | $P_{6,2}$ | Spectral | 1 | 0.2988 | 0.8815 | 27.88 | 1074 |
| 232 | $P_{6,4}$ | VQC | 0 | 0.1939 | 0.8785 | 3752.97 | 1037 |
| 236 | $P_{6,4}$ | VQC | 1 | 0.1376 | 0.5919 | 5044.49 | 1030 |
| 205 | $P_{6,4}$ | MLP | 0 | 0.1959 | 0.3409 | 15.61 | 1031 |
| 210 | $P_{6,4}$ | MLP | 1 | 0.0844 | 0.3872 | 15.28 | 1045 |
| 181 | $P_{6,4}$ | Spectral | 0 | 0.1650 | 0.7594 | 33.29 | 1046 |
| 182 | $P_{6,4}$ | Spectral | 1 | 0.2667 | 0.8056 | 27.02 | 1035 |
| 237 | $P_{6,8}$ | VQC | 0 | 0.0606 | 0.3951 | 8682.91 | 1028 |
| 238 | $P_{6,8}$ | VQC | 1 | 0.0591 | 0.4977 | 11551.89 | 1035 |
| 211 | $P_{6,8}$ | MLP | 0 | 0.0359 | 0.0935 | 22.42 | 1038 |
| 212 | $P_{6,8}$ | MLP | 1 | 0.0756 | 0.1455 | 19.10 | 1027 |
| 183 | $P_{6,8}$ | Spectral | 0 | 0.1327 | 0.3583 | 34.25 | 1048 |
| 184 | $P_{6,8}$ | Spectral | 1 | 0.1513 | 0.4064 | 30.21 | 1059 |
| 239 | $P_{8,2}$ | VQC | 0 | 0.6694 | 1.7132 | 8575.21 | 1097 |
| 240 | $P_{8,2}$ | VQC | 1 | 1.0819 | 2.3649 | 11564.26 | 1089 |
| 213 | $P_{8,2}$ | MLP | 0 | 0.1100 | 0.2196 | 16.58 | 1046 |

Continued on next page

| Index | Design | Model | Seed | $e_T$ | $e_\sigma$ | Time (s) | Closures |
|---|---|---|---|---|---|---|---|
| 214 | $P_{8,2}$ | MLP | 1 | 0.1005 | 0.2610 | 14.70 | 1044 |
| 185 | $P_{8,2}$ | Spectral | 0 | 0.3864 | 0.6899 | 29.84 | 1057 |
| 186 | $P_{8,2}$ | Spectral | 1 | 0.1796 | 0.4204 | 25.73 | 1037 |
| 231 | $C_{64}$ | VQC | 0 | 0.1051 | 0.8078 | 1148.63 | 1032 |
| 235 | $C_{64}$ | VQC | 1 | 0.3348 | 0.8318 | 1215.85 | 1037 |
| 204 | $C_{64}$ | MLP | 0 | 0.0870 | 0.2023 | 9.31 | 1027 |
| 209 | $C_{64}$ | MLP | 1 | 0.1705 | 0.2060 | 8.90 | 1042 |
| 230 | $C_{256}$ | VQC | 0 | 0.2140 | 0.5376 | 2905.75 | 1021 |
| 234 | $C_{256}$ | VQC | 1 | 0.2176 | 0.7998 | 3397.30 | 1069 |
| 202 | $C_{256}$ | MLP | 0 | 0.0906 | 0.2438 | 11.66 | 1037 |
| 207 | $C_{256}$ | MLP | 1 | 0.1491 | 0.2713 | 11.04 | 1034 |
| 229 | $C_{1024}$ | VQC | 0 | 0.1815 | 0.8250 | 10380.71 | 1040 |
| 233 | $C_{1024}$ | VQC | 1 | 0.2463 | 0.6938 | 9245.77 | 1050 |
| 201 | $C_{1024}$ | MLP | 0 | 0.0770 | 0.0886 | 21.79 | 1056 |
| 206 | $C_{1024}$ | MLP | 1 | 0.1154 | 0.2566 | 19.22 | 1043 |
| 203 | $C_{4096}$ | MLP | 0 | 0.1118 | 0.3441 | 73.53 | 1055 |
| 208 | $C_{4096}$ | MLP | 1 | 0.1213 | 0.2496 | 56.15 | 1032 |
| 216 | $D_{16}$ | VQC | 0 | 0.9935 | 2.5130 | 28.95 | 0 |
| 220 | $D_{16}$ | VQC | 1 | 2.2859 | 1.7633 | 27.89 | 0 |
| 188 | $D_{16}$ | MLP | 0 | 0.7236 | 1.7778 | 0.66 | 0 |
| 192 | $D_{16}$ | MLP | 1 | 1.4504 | 1.2370 | 0.57 | 0 |
| 218 | $D_{64}$ | VQC | 0 | 0.2571 | 0.6260 | 32.40 | 0 |
| 222 | $D_{64}$ | VQC | 1 | 0.3024 | 0.4412 | 31.38 | 0 |
| 190 | $D_{64}$ | MLP | 0 | 0.0865 | 0.2229 | 0.70 | 0 |
| 194 | $D_{64}$ | MLP | 1 | 0.0738 | 0.1577 | 0.63 | 0 |
| 217 | $D_{256}$ | VQC | 0 | 0.1873 | 0.2953 | 49.46 | 0 |
| 221 | $D_{256}$ | VQC | 1 | 0.1940 | 0.2882 | 46.74 | 0 |
| 189 | $D_{256}$ | MLP | 0 | 0.0873 | 0.0773 | 0.77 | 0 |
| 193 | $D_{256}$ | MLP | 1 | 0.1140 | 0.0995 | 0.64 | 0 |
| 215 | $D_{1024}$ | VQC | 0 | 0.1769 | 0.2211 | 156.30 | 0 |
| 219 | $D_{1024}$ | VQC | 1 | 0.2070 | 0.2146 | 141.83 | 0 |
| 187 | $D_{1024}$ | MLP | 0 | 0.0870 | 0.0885 | 1.19 | 0 |
| 191 | $D_{1024}$ | MLP | 1 | 0.0682 | 0.0858 | 0.93 | 0 |

## S9.2 Secondary wave speeds: all executed runs

The 24 secondary-speed records use the exponential pulse, seeds $0, 1, 2$, $N_c = 512$, 2048 fixed L-BFGS points, 2000 Adam steps, and a cap of 1000 L-BFGS iterations. Counts and deviations are the $(6, 4)$ entries in Table S2; the unmatched reference has 12802 parameters. Rates are 0.1, except 0.01 for the reference. The main secondary-wave-speed table summarizes the errors with sample standard deviations; the $\gamma = 9$ baseline there uses only E1 seeds $0, 1, 2$. Table S17 gives the new records without conflating their measured times with an isolated resource cost.

Table S17: Secondary-wave-speed per-seed results. Index refers to the zero-based position in `runs_all.json`. Closures are actual objective/gradient evaluations, not optimizer iterations.

| Index | $\gamma$ | Model | Seed | $e_T$ | $e_\sigma$ | Time (s) | Closures |
|---|---|---|---|---|---|---|---|
| 259 | 25 | VQC | 0 | 0.3175 | 1.7249 | 4093.61 | 1029 |
| 260 | 25 | VQC | 1 | 0.2399 | 0.8438 | 4111.89 | 1036 |
| 261 | 25 | VQC | 2 | 0.3208 | 1.0225 | 4142.69 | 1038 |
| 247 | 25 | MLP | 0 | 0.1595 | 0.2019 | 21.92 | 1052 |
| 248 | 25 | MLP | 1 | 0.0674 | 0.1023 | 20.29 | 1031 |
| 249 | 25 | MLP | 2 | 0.1569 | 0.4017 | 20.30 | 1064 |
| 241 | 25 | Spectral | 0 | 0.3349 | 0.5232 | 61.63 | 1042 |
| 242 | 25 | Spectral | 1 | 0.2136 | 1.0056 | 52.32 | 1037 |
| 243 | 25 | Spectral | 2 | 0.4054 | 0.3738 | 55.55 | 1050 |
| 253 | 25 | Reference | 0 | 0.0221 | 0.1237 | 239.86 | 1074 |
| 254 | 25 | Reference | 1 | 0.0193 | 0.0632 | 239.85 | 1059 |
| 255 | 25 | Reference | 2 | 0.0250 | 0.1264 | 238.74 | 1059 |

Continued on next page

| Index | $\gamma$ | Model | Seed | $e_T$ | $e_\sigma$ | Time (s) | Closures |
|---|---|---|---|---|---|---|---|
| 262 | 49 | VQC | 0 | 0.3257 | 2.3413 | 4112.72 | 1028 |
| 263 | 49 | VQC | 1 | 0.5982 | 1.2869 | 6171.44 | 1044 |
| 264 | 49 | VQC | 2 | 0.2366 | 0.7172 | 6078.08 | 1037 |
| 250 | 49 | MLP | 0 | 0.1962 | 0.3184 | 19.53 | 1036 |
| 251 | 49 | MLP | 1 | 0.2281 | 0.5806 | 18.53 | 1034 |
| 252 | 49 | MLP | 2 | 0.3161 | 0.4543 | 18.16 | 1065 |
| 244 | 49 | Spectral | 0 | 0.6394 | 0.9249 | 55.14 | 1062 |
| 245 | 49 | Spectral | 1 | 0.2615 | 0.5832 | 42.71 | 1055 |
| 246 | 49 | Spectral | 2 | 0.4190 | 0.5600 | 44.30 | 1062 |
| 256 | 49 | Reference | 0 | 0.0436 | 0.1029 | 228.53 | 1056 |
| 257 | 49 | Reference | 1 | 0.0417 | 0.1117 | 208.67 | 1047 |
| 258 | 49 | Reference | 2 | 0.0408 | 0.0721 | 204.36 | 1056 |

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